



DEPARTMENT OF MATHEMATICS

Sub Code:	AI34/AD34/CI35/CY35	Sub:	Discrete Mathematical Structures	Sec:	AI/AD/CI/CY
Semester:	III	Term:	03-10-2024 to 25-01-2025	Marks:	30
Date:	29-11-2024	Time:	3:00 PM-4:00 PM	Test:	I

Note: Answer any TWO full questions. Each main question carries 15 marks

Q. No.	Questions	Blooms Level	CO's	Marks
1. (a)	Let $A = \{a, b, c, d\}$ and R be a relation on A that has the matrix $M_R = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$ Construct the digraph of R and list the in-degrees and out-degrees of all vertices.	CO2	L3	5
(b)	Let $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4\}$. The relations R and S from A to B are given by the matrix of relations $M_R = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$ $M_S = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix}$ Determine the relations $\bar{R}, \bar{S}, R \cup S, R \cap S, R^{-1}, S^{-1}$.	CO2	L4	5
(c)	Prove that If $n = ab$ where a, b are positive integers then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$ by contrapositive method.	CO1	L5	5
2. (a)	Let $A = \{1, 2, 3, 4, 6, 12\}$ and on A define the relation R by aRb if a divides b . Prove that R is a POSET on A and draw the Hasse diagram of this relation.	CO2	L3	5
(b)	Prove that $11^n - 4^n$ is divisible by 7, $\forall n \in \mathbb{Z}^+$ by mathematical induction.	CO1	L4	5
(c)	Using truth table, verify the compound statement $[(p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow [(p \vee q) \rightarrow r]$ is a tautology or not?	CO1	L5	5
3. (a)	Write the converse, inverse and contrapositive for if a quadrilateral is a parallelogram then its diagonals bisect each other.	CO1	L3	5
(b)	If R is a relation on $\mathbb{Z}$, the set of all integers such that xRy iff $x - y$ is divisible by 5. Verify that R is an equivalence relation or not?	CO2	L4	5
(c)	Without constructing truth table prove that $(p \vee q) \wedge (\neg p \wedge (\neg p \wedge q))$ is logically equivalent to $\neg(p \wedge q)$.	CO1	L5	5